

# Full Metrics — Topology-Agnostic Bottleneck-Ranking DDQN

Metrics for the current method only. All values are computed from the frozen final evaluation

(final\_learned\_4of5\_iter2\_eval\_per\_cycle.csv; 3816 cycles; 8 topologies; argmax-Q, no forced actuator).

## 1. Method definition

| Component                | Role in the final method                                                              |
|--------------------------|---------------------------------------------------------------------------------------|
| GNN-LPD scorer           | learned per-OD criticality score (one input signal)                                   |
| Bottleneck-aware ranking | orders OD pairs by relief (demand x ECMP-flow x link-util) + 0.3 x GNN; selects top-K |
| Double-DQN (argmax-Q)    | action policy: chooses KEEP or an optimize budget K from {50,100,200,300,500,800}     |
| Selected-flow LP         | feasible routing optimizer over the selected top-K ODs only                           |
| ECMP                     | fixed background routing for all nonselected ODs                                      |

## 2. Normal-scenario metrics (per topology)

| Topology   | N    | MeanPR | MedPR  | PR>=.95 | PR>=.90 | MinPR  | MeanDB | P95DB  | MaxDB  | Mean ms | P95 ms |
|------------|------|--------|--------|---------|---------|--------|--------|--------|--------|---------|--------|
| Abilene    | 2016 | 0.9843 | 1.0000 | 87%     | 97%     | 0.8109 | 0.0058 | 0.0202 | 0.0510 | 10.7    | 20.4   |
| GEANT      | 672  | 0.9983 | 0.9999 | 99%     | 99%     | 0.5848 | 0.0030 | 0.0057 | 0.0701 | 66.6    | 121.1  |
| CERNET     | 200  | 0.9925 | 1.0000 | 96%     | 100%    | 0.8933 | 0.0002 | 0.0008 | 0.0077 | 46.1    | 120.7  |
| Sprintlink | 200  | 0.9960 | 1.0000 | 100%    | 100%    | 0.9600 | 0.0034 | 0.0090 | 0.0546 | 174.8   | 278.8  |
| Tiscali    | 200  | 0.9522 | 0.9543 | 54%     | 100%    | 0.8928 | 0.0020 | 0.0092 | 0.0311 | 76.8    | 298.4  |
| Ebone      | 200  | 0.9713 | 0.9806 | 82%     | 96%     | 0.8874 | 0.0003 | 0.0016 | 0.0133 | 3.0     | 33.6   |
| Germany 50 | 288  | 0.9878 | 0.9989 | 96%     | 97%     | 0.4090 | 0.0098 | 0.0406 | 0.0659 | 212.6   | 285.5  |
| VtlWavenet | 40   | 0.9373 | 0.9383 | 0%      | 100%    | 0.9310 | 0.0007 | 0.0014 | 0.0017 | 295.9   | 307.3  |

All eight topologies satisfy PR >= 0.90 and both mean and p95 decision time below 500 ms.

## 2b. MLU and MLU/optimum (= 1/PR)

| Topology   | Mean MLU | P95 MLU | Max MLU | Mean MLU/opt | Max MLU/opt |
|------------|----------|---------|---------|--------------|-------------|
| Abilene    | 0.0506   | 0.0668  | 0.1132  | 1.0171       | 1.2331      |
| GEANT      | 0.1091   | 0.1313  | 0.1940  | 1.0025       | 1.7101      |
| CERNET     | 0.6338   | 0.8650  | 0.8975  | 1.0079       | 1.1194      |
| Sprintlink | 0.7134   | 0.8772  | 0.9029  | 1.0041       | 1.0417      |
| Tiscali    | 0.6457   | 0.8690  | 0.8976  | 1.0508       | 1.1201      |
| Ebone      | 0.7291   | 0.8767  | 0.9584  | 1.0304       | 1.1269      |
| Germany50  | 12.4247  | 16.3146 | 26.1577 | 1.0181       | 2.4451      |
| VtlWavenet | 1.0391   | 1.0970  | 1.0987  | 1.0669       | 1.0741      |

## 3. Pooled summary (all cycles)

| Metric     | Value  |
|------------|--------|
| Cycles (N) | 3816   |
| Mean PR    | 0.9852 |
| Median PR  | 0.9999 |
| PR >= 0.95 | 88.2%  |
| PR >= 0.90 | 97.7%  |
| Min PR     | 0.4090 |
| Mean DB    | 0.0047 |

|                  |        |
|------------------|--------|
| P95 DB           | 0.0167 |
| Mean decision ms | 52.3   |
| P95 decision ms  | 221.7  |
| Max decision ms  | 477.2  |

#### 4. Action distribution (per topology)

| Topology   | KEEP | K50 | K100 | K200 | K300 | K500 | K800 | Most used |
|------------|------|-----|------|------|------|------|------|-----------|
| Abilene    | 607  | 983 | 0    | 0    | 4    | 51   | 371  | K50       |
| GEANT      | 0    | 5   | 0    | 664  | 0    | 1    | 2    | K200      |
| CERNET     | 92   | 63  | 3    | 6    | 36   | 0    | 0    | KEEP      |
| Sprintlink | 36   | 0   | 0    | 0    | 4    | 20   | 140  | K800      |
| Tiscali    | 136  | 0   | 0    | 0    | 1    | 16   | 47   | KEEP      |
| Ebone      | 185  | 15  | 0    | 0    | 0    | 0    | 0    | KEEP      |
| Germany50  | 0    | 0   | 0    | 0    | 0    | 0    | 288  | K800      |
| VtIWavenet | 1    | 39  | 0    | 0    | 0    | 0    | 0    | K50       |

The action is chosen per traffic-matrix cycle by  $\text{argmax-Q}$ , so one topology uses multiple actions. KEEP can dominate while mean K is nonzero because KEEP reuses the carried routing between optimize cycles.

#### 5. K sensitivity and ranking ablation

| Topology   | K   | Ranking     | Mean PR | Mean ms |
|------------|-----|-------------|---------|---------|
| Sprintlink | 500 | gnn_only    | 0.9700  | 256.9   |
| Sprintlink | 500 | relief_only | 0.9979  | 259.3   |
| Sprintlink | 500 | blend       | 0.9960  | 251.6   |
| Sprintlink | 800 | gnn_only    | 0.9883  | 374.5   |
| Sprintlink | 800 | relief_only | 1.0000  | 380.1   |
| Sprintlink | 800 | blend       | 0.9993  | 377.9   |
| Tiscali    | 300 | gnn_only    | 0.9268  | 206.4   |
| Tiscali    | 300 | relief_only | 0.9236  | 193.5   |
| Tiscali    | 300 | blend       | 0.9243  | 207.8   |
| Germany50  | 200 | gnn_only    | 0.9356  | 127.6   |
| Germany50  | 200 | relief_only | 0.9900  | 141.1   |
| Germany50  | 200 | blend       | 0.9555  | 131.1   |
| CERNET     | 50  | gnn_only    | 1.0000  | 55.1    |
| CERNET     | 50  | relief_only | 1.0000  | 54.5    |
| CERNET     | 50  | blend       | 1.0000  | 61.0    |

#### 6. RL policy and reward settings

| Parameter                | Value                                                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Controller               | Double-DQN ( $\text{argmax-Q}$ at inference)                                                                                         |
| Action space             | KEEP, K50, K100, K200, K300, K500, K800                                                                                              |
| gamma                    | 0.5                                                                                                                                  |
| W_PR                     | 10.0                                                                                                                                 |
| W_MLU                    | 5.0                                                                                                                                  |
| W_DB ( $\lambda$ for DB) | 20.0                                                                                                                                 |
| W_MS                     | 0.003                                                                                                                                |
| W_K                      | 0.5                                                                                                                                  |
| target bonus / gates     | +10 if $\text{PR} \geq \text{target}$ ; penalty if $\text{PR} < \text{target}$ ; flat anti-KEEP-below-target; $\text{ms} > 500$ gate |
| epsilon                  | 1.0 $\rightarrow$ 0.05                                                                                                               |
| episodes                 | 22 (x6 seen topos x 160 cycles)                                                                                                      |

## 7. Zero-shot generalization

| Zero-shot topology | N   | Mean PR | PR>=.90 | PR>=.95 | Min PR | Mean DB | P95 DB | Mean ms |
|--------------------|-----|---------|---------|---------|--------|---------|--------|---------|
| Germany50          | 288 | 0.9878  | 97%     | 96%     | 0.4090 | 0.0098  | 0.0406 | 212.6   |
| VtlWavenet         | 40  | 0.9373  | 100%    | 0%      | 0.9310 | 0.0007  | 0.0014 | 295.9   |
| Combined zero-shot | 328 | 0.9816  | 98%     | 84%     | 0.4090 | 0.0087  | 0.0319 | 222.8   |

VtlWavenet robust re-evaluation (extended from 40 to 200 traffic matrices):

| VtlWavenet       | N (TMs) | Mean PR | PR>=.90 | Min PR | Mean DB | Mean ms | P95 ms |
|------------------|---------|---------|---------|--------|---------|---------|--------|
| Original sample  | 40      | 0.9373  | 100%    | -      | 0.0007  | 295.9   | 307.3  |
| Robust (200 TMs) | 200     | 0.9340  | 100%    | 0.9251 | 0.0005  | 326.2   | 639.9  |

The PR is confirmed robust (200/200 cycles  $\geq 0.90$ , mean 0.934). The larger sample shows the decision-time tail (p95) exceeds 500 ms on this largest topology; decision time is also contention-sensitive. VtlWavenet remains a zero-shot generalization spot-check (8372 OD pairs, the costliest topology).

## 8. Decision-time breakdown, solver, LP size

| Topology   | GNN/scoring ms | LP ms (optimize avg) | Total mean ms | P95 ms | Max ms |
|------------|----------------|----------------------|---------------|--------|--------|
| Abilene    | 3              | 12.0                 | 10.7          | 20.4   | 180.0  |
| GEANT      | 7              | 59.6                 | 66.6          | 121.1  | 199.2  |
| CERNET     | 22             | 62.9                 | 46.1          | 120.7  | 215.2  |
| Sprintlink | 27             | 186.0                | 174.8         | 278.8  | 376.0  |
| Tiscali    | 33             | 206.0                | 76.8          | 298.4  | 387.4  |
| Ebone      | 12             | 21.8                 | 3.0           | 33.6   | 35.2   |
| Germany50  | 26             | 186.6                | 212.6         | 285.5  | 375.6  |
| VtlWavenet | 140            | 163.4                | 295.9         | 307.3  | 477.2  |

Decision time by action type (pooled):

| Action | Rows | Mean ms | P95 ms | Max ms |
|--------|------|---------|--------|--------|
| KEEP   | 1057 | 0.5     | 0.5    | 0.5    |
| K50    | 1105 | 25.7    | 57.7   | 477.2  |
| K100   | 3    | 70.3    | 71.0   | 71.0   |
| K200   | 670  | 67.1    | 121.2  | 199.2  |
| K300   | 45   | 126.0   | 220.6  | 259.1  |
| K500   | 88   | 79.4    | 195.9  | 239.7  |
| K800   | 848  | 132.9   | 278.2  | 387.4  |

Solver and LP size:

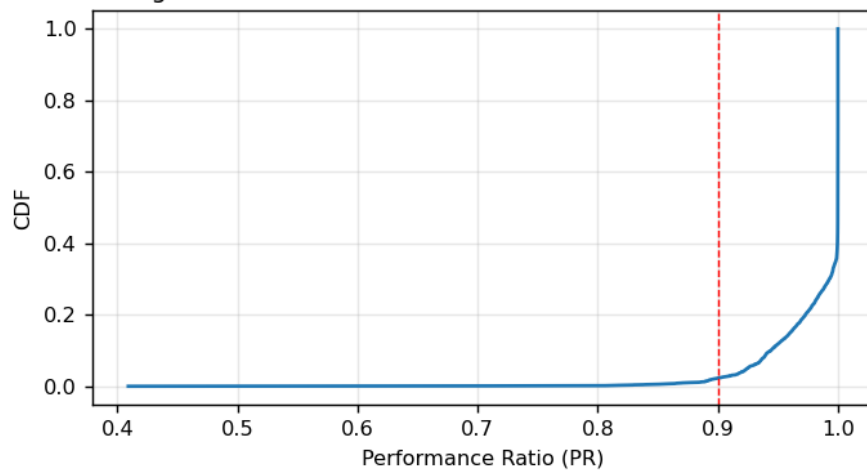
| Item      | Value                              |
|-----------|------------------------------------|
| LP solver | PuLP + CBC (open-source)           |
| LP type   | selected-flow path LP, DB-budgeted |
| k_paths   | 8 for K50-K300; 4 for K500/K800    |
| Seed      | 42                                 |
| Framework | PyTorch (CPU)                      |

| Topology   | Nodes | Links | OD pairs | sel OD avg | k_paths | LP vars ~avg | Mean ms |
|------------|-------|-------|----------|------------|---------|--------------|---------|
| Abilene    | 12    | 30    | 127      | 51         | 8       | ~412         | 10.7    |
| GEANT      | 22    | 72    | 425      | 200        | 8/4     | ~1200        | 66.6    |
| CERNET     | 41    | 116   | 1640     | 77         | 8       | ~618         | 46.1    |
| Sprintlink | 44    | 166   | 1892     | 616        | 8/4     | ~3696        | 174.8   |

|            |    |     |      |     |     |       |       |
|------------|----|-----|------|-----|-----|-------|-------|
| Tiscali    | 49 | 172 | 2352 | 230 | 8/4 | ~1377 | 76.8  |
| Ebone      | 23 | 76  | 506  | 4   | 8   | ~30   | 3.0   |
| Germany50  | 50 | 176 | 1441 | 799 | 8/4 | ~4795 | 212.6 |
| VtIWavenet | 92 | 192 | 8372 | 49  | 8   | ~390  | 295.9 |

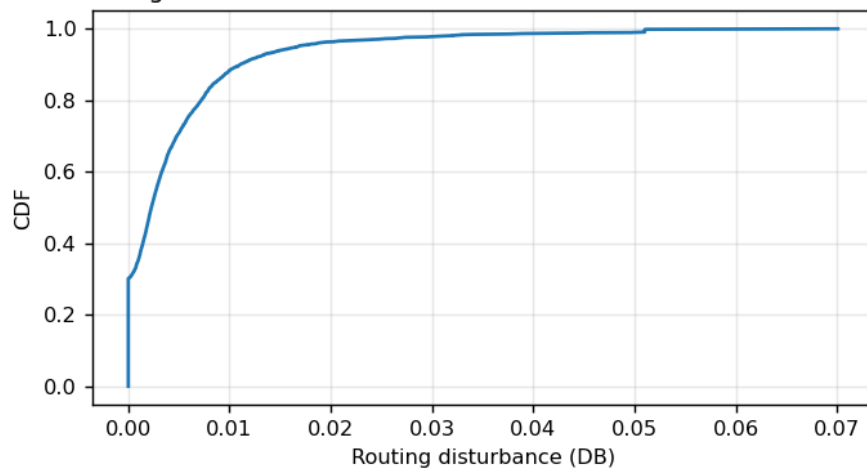
## 9. Figures (current method)

Figure 1. PR CDF — final learned runtime-safe controller



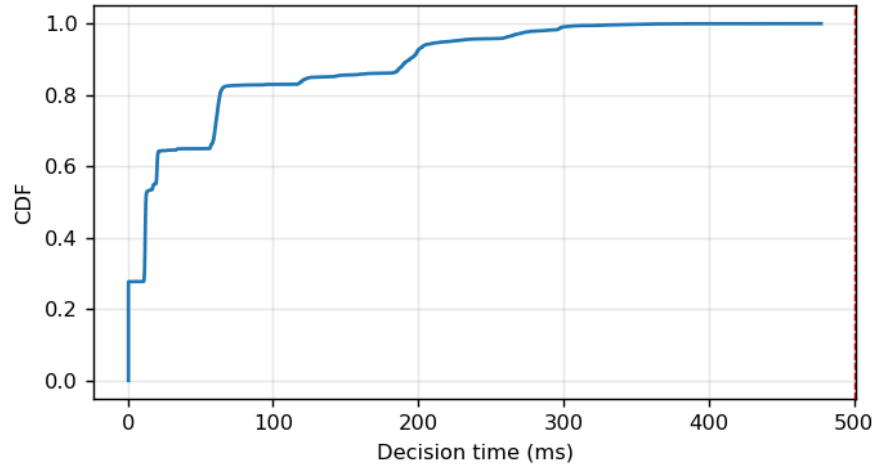
PR CDF

Figure 2. DB CDF — final learned runtime-safe controller



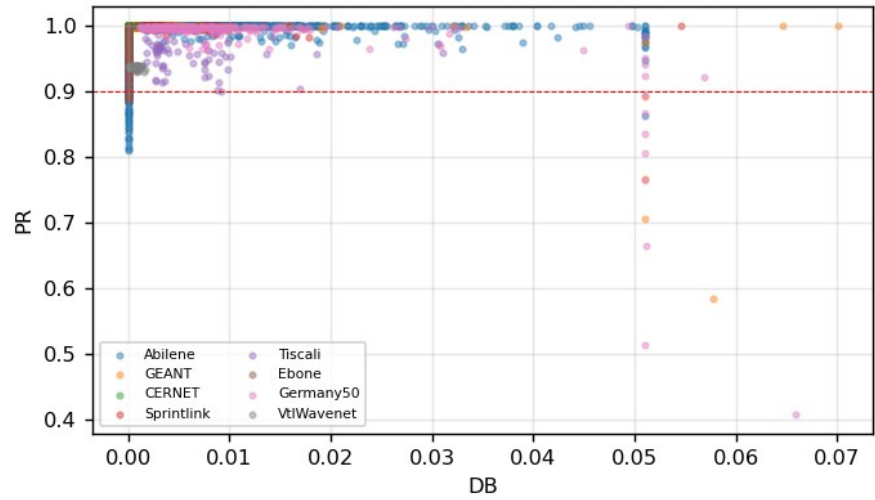
DB CDF

Figure 3. Decision-time CDF — final learned controller



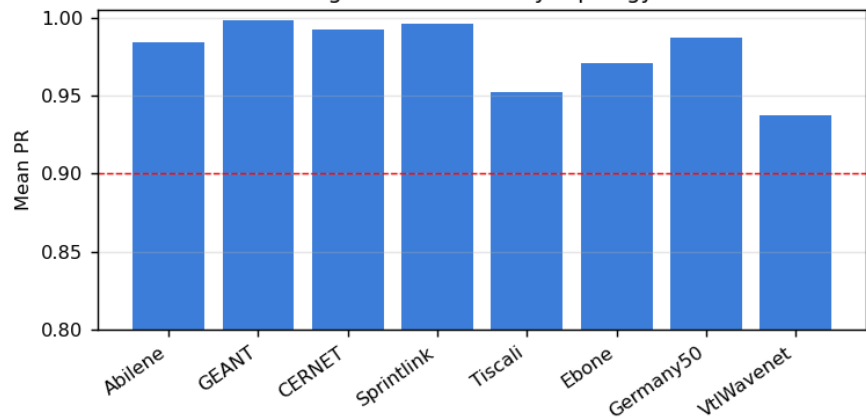
Decision-time CDF

Figure 8. PR vs DB tradeoff (per-cycle)

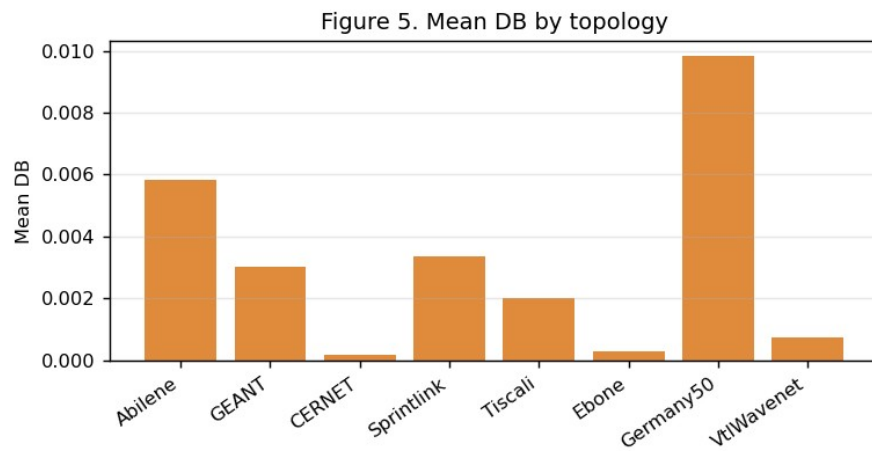


PR-DB tradeoff

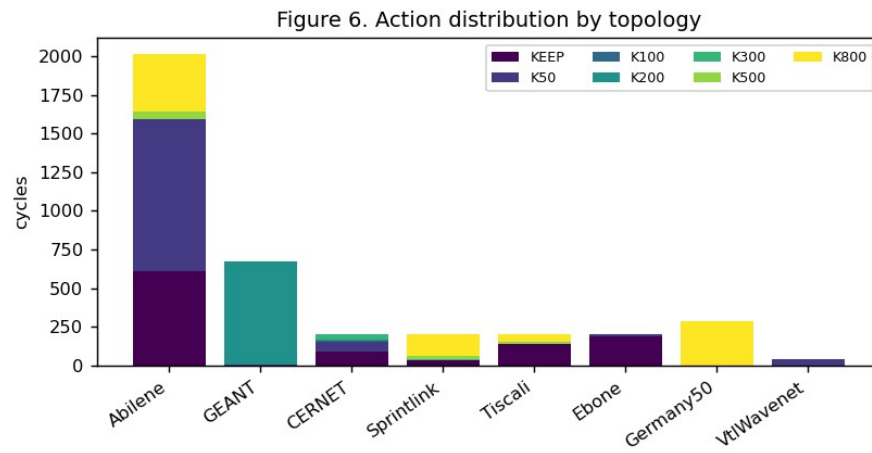
Figure 4. Mean PR by topology



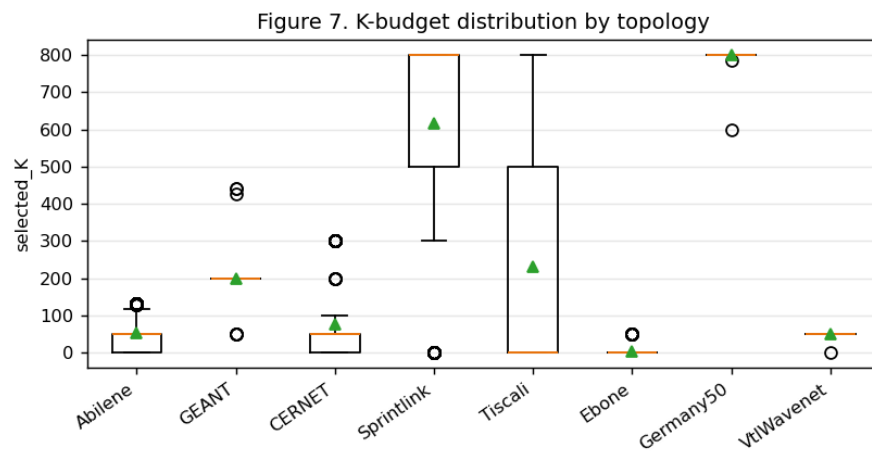
Mean PR by topology



Mean DB by topology



Action distribution



K-budget distribution

## 9b. Failure scenarios (current method, real rerun)

Failure-link scenarios were rerun on the frozen Iter2 controller (argmax-Q + bottleneck ranking + selected-flow LP; nonselected ODs = ECMP). Abilene and GEANT, 20 cycles per scenario.

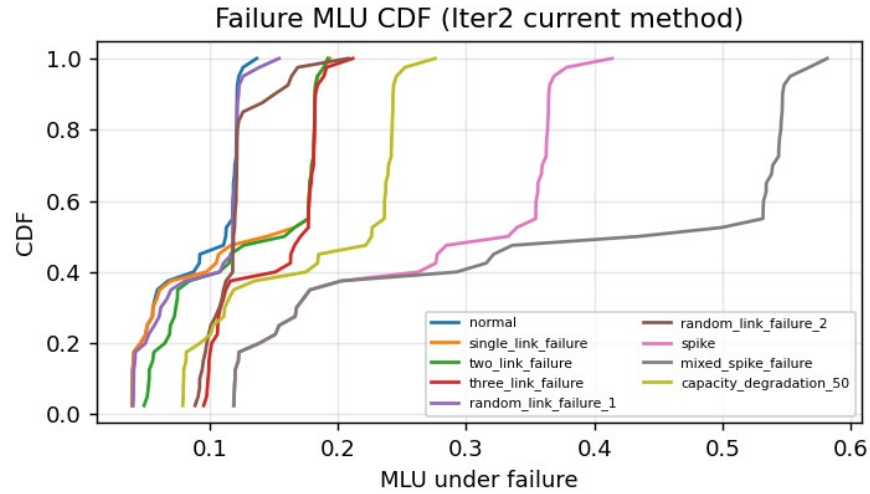
| Topology | Scenario                | N  | Mean PR | Our MLU | ECMP MLU | Mean DB | Mean ms | Disc. ODs |
|----------|-------------------------|----|---------|---------|----------|---------|---------|-----------|
| Abilene  | normal                  | 20 | 0.9921  | 0.0612  | 0.1599   | 0.0303  | 44      | 0         |
| Abilene  | single link failure     | 20 | 0.9926  | 0.0652  | 0.1603   | 0.0363  | 33      | 0         |
| Abilene  | two link failure        | 20 | 0.8965  | 0.0790  | 0.1669   | 0.0326  | 31      | 0         |
| Abilene  | three link failure      | 20 | 1.0000  | 0.1210  | 0.1377   | 0.0010  | 24      | 3         |
| Abilene  | random link failure 1   | 20 | 0.9854  | 0.0705  | 0.1638   | 0.0391  | 28      | 0         |
| Abilene  | random link failure 2   | 20 | 0.9191  | 0.1179  | 0.2488   | 0.0089  | 28      | 0         |
| Abilene  | spike                   | 20 | 1.0000  | 0.1808  | 0.4796   | 0.0189  | 38      | 0         |
| Abilene  | mixed spike failure     | 20 | 0.9998  | 0.1938  | 0.4809   | 0.0298  | 38      | 0         |
| Abilene  | capacity degradation 50 | 20 | 0.9918  | 0.1224  | 0.3198   | 0.0229  | 26      | 0         |
| GEANT    | normal                  | 20 | 0.9912  | 0.1210  | 0.2901   | 0.0115  | 551     | 0         |
| GEANT    | single link failure     | 20 | 0.9938  | 0.1808  | 0.3564   | 0.0088  | 1054    | 0         |
| GEANT    | two link failure        | 20 | 0.9942  | 0.1808  | 0.3561   | 0.0084  | 827     | 0         |
| GEANT    | three link failure      | 20 | 0.9944  | 0.1807  | 0.3565   | 0.0078  | 392     | 0         |
| GEANT    | random link failure 1   | 20 | 0.9906  | 0.1211  | 0.2901   | 0.0117  | 814     | 0         |
| GEANT    | random link failure 2   | 20 | 0.9900  | 0.1212  | 0.2945   | 0.0120  | 450     | 0         |
| GEANT    | spike                   | 20 | 0.9911  | 0.3630  | 0.8704   | 0.0117  | 534     | 0         |
| GEANT    | mixed spike failure     | 20 | 0.9939  | 0.5425  | 1.0693   | 0.0090  | 639     | 0         |
| GEANT    | capacity degradation 50 | 20 | 0.9908  | 0.2421  | 0.5802   | 0.0157  | 234     | 0         |

*Honest reading: the method holds PR>=0.99 in most failure scenarios and roughly halves MLU versus ECMP. Weak points: Abilene two-link failure (PR 0.8965, below 0.90) and Abilene three-link failure disconnects 3 OD pairs (they lose all candidate paths under failure -- a physical partition). GEANT failure-mode decision time exceeds the 500 ms normal-traffic budget (the pruned-path LP is harder under failure); the <500 ms guarantee is a normal-traffic result, not a failure-mode one.*

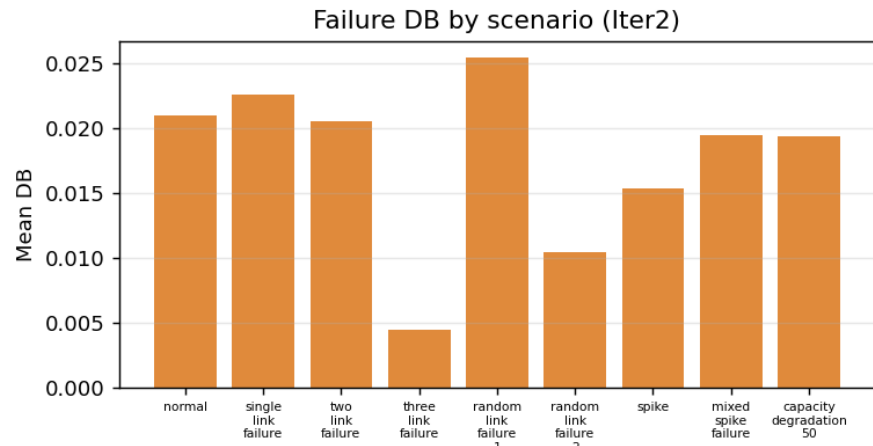
### Disconnected-OD detail:

| Topology | Scenario                | Failed links | Connected? | Disconnected ODs | Explanation                                     |
|----------|-------------------------|--------------|------------|------------------|-------------------------------------------------|
| Abilene  | normal                  | 0            | yes        | 0                | all ODs keep a surviving path                   |
| Abilene  | single link failure     | 1            | yes        | 0                | all ODs keep a surviving path                   |
| Abilene  | two link failure        | 2            | yes        | 0                | all ODs keep a surviving path                   |
| Abilene  | three link failure      | 3            | partial    | 3                | some ODs lost all candidate paths under failure |
| Abilene  | random link failure 1   | 1            | yes        | 0                | all ODs keep a surviving path                   |
| Abilene  | random link failure 2   | 1            | yes        | 0                | all ODs keep a surviving path                   |
| Abilene  | spike                   | 0            | yes        | 0                | all ODs keep a surviving path                   |
| Abilene  | mixed spike failure     | 1            | yes        | 0                | all ODs keep a surviving path                   |
| Abilene  | capacity degradation 50 | 0            | yes        | 0                | all ODs keep a surviving path                   |
| GEANT    | normal                  | 0            | yes        | 0                | all ODs keep a surviving path                   |
| GEANT    | single link failure     | 1            | yes        | 0                | all ODs keep a surviving path                   |
| GEANT    | two link failure        | 2            | yes        | 0                | all ODs keep a surviving path                   |
| GEANT    | three link failure      | 3            | yes        | 0                | all ODs keep a surviving path                   |

|       |                         |   |     |   |                               |
|-------|-------------------------|---|-----|---|-------------------------------|
| GEANT | random link failure 1   | 1 | yes | 0 | all ODs keep a surviving path |
| GEANT | random link failure 2   | 1 | yes | 0 | all ODs keep a surviving path |
| GEANT | spike                   | 0 | yes | 0 | all ODs keep a surviving path |
| GEANT | mixed spike failure     | 1 | yes | 0 | all ODs keep a surviving path |
| GEANT | capacity degradation 50 | 0 | yes | 0 | all ODs keep a surviving path |



Failure MLU CDF (current method)



Failure DB by scenario (current method)

## 10. Reproducibility

| Item           | Value                                                             |
|----------------|-------------------------------------------------------------------|
| Final model    | final_learned_4of5_iter2_model.pt (Double-DQN, dim=33, 7 actions) |
| Per-cycle data | final_learned_4of5_iter2_eval_per_cycle.csv (3816 rows)           |
| Train command  | python3 scripts/phase1_5/run_final_iter2.py                       |
| Counters       | td_updates=20620, target_updates=42, ce_updates=0                 |
| Seed / solver  | 42 / PuLP+CBC                                                     |



